Applied NLP

DATA 641

Lecture 15 — Recommender Systems for Textual Data

*Practical Natural Language Processing: A Comprehensive Guide to Building Real-World NLP Systems 1st Edition, Sowmya Vajjala, Bodhisattwa Majumder, Anuj Gupta, Harshit Surana, O' Reilly and Natural Language Processing in Action, Understanding, analyzing, and generating text with Python, Hobson Lane, Cole Howard, Hannes Hapke, First edition, MANNING

Recommender Systems

A recommender system, also known as a recommendation system or recommender engine, is a software application or algorithm that provides personalized suggestions or recommendations to users. A popular approach to build a recommender system is called collaborative filtering:

- Collaborative filtering is a prevalent approach in building recommendation systems. It recommends items or content to users based on their past behavior and preferences. Recommends items that users with similar profiles have liked or interacted with in the past.

- Netflix utilizes collaborative filtering at a large scale for its recommendation engine. Users receive movie and show recommendations based on their viewing history and preferences, as well as what similar users have enjoyed.

Content-Based Recommendation Systems

- An example of one such recommendation is the "related articles" feature on newspaper websites.
- One approach to building such a content-based recommendation system is to use a topic model like we saw in this module. Texts similar to the current text in terms of topic distribution can be shown as "related" texts.

"This book can be for non-black people in Canada who understand that there is a struggle for black liberation in this country — whether they use that language or not. It's for those readers who see demonstrations on the street that are explicitly about blackness by black people and are curious and who want to understand. *The Skin We're In* exists so those who are not black can better articulate the struggle that they see or hear about — so that they can better understand and contextualize it.

"I had the great privilege and honour of taking the time to find all of these things and put them together for the book."

- **7 works of Canadian nonfiction to read for Black History Month 2020**

Desmond Cole's comments have been edited for length and clarity.

RELATED STORIES

- **Six Canadian writers of black heritage to watch in 2020**
- **Why Desmond Cole says empathy isn't enough to eradicate anti-black racism**
- **40 works of Canadian nonfiction to watch for in spring 2020**

Practical advices

Evaluating Recommendation Systems:

- Performance Indicators. Measure the impact of recommendations using performance indicators:
    - User click-through rates.
    - Conversion into a purchase (if applicable).
    - Customer engagement on the website.
- A/B Testing:
    - Conduct A/B tests to compare different recommendation approaches.
    - Different user groups are exposed to varied recommendations.
    - Performance indicators are analyzed to assess the effectiveness of recommendations.

- User Studies:
    - Perform user studies with carefully designed experiments.
    - Participants are shown specific recommendations and asked to rate them for evaluation.
- Test Set Comparison:
    - Use a small test set with appropriate recommendations for a specific item.
    - Evaluate the recommendation system by comparing its output to this test set.
- Combination of Indicators:
    - Industry-scale recommendation systems often combine multiple evaluation indicators.
    - Analytics platforms like Google Analytics are valuable for monitoring and analyzing system performance.

Practical advices

Pre-processing Decisions:

- Significance of Pre-processing:
  - Pre-processing decisions significantly impact the recommendations generated by the system.
  - Pre-processing involves steps like tokenization, lowercasing, special character removal, and more.
- Knowing Objectives:
  - Define clear objectives before deciding on a pre-processing approach.
  - Consider the specific requirements of the recommendation system.